

UCSD / Blum Summer Field Internship

2019 Final Report: Tijuana Estuary, US – Divina Providencia, MX



2019 group in Downtown Tijuana.

Description:

This report details the sixth installment of the UCSD Blum Summer Field Internship at the UCSD Cross-Border Community Station in Divina Providencia, Tijuana. Interns are selected from a talented pool of applicants across various disciplines and academic programs at UCSD and are asked to do a mixture of academic, service, and community engagement work. New interns take a class at UCSD and visit partner organizations in the spring quarter, before being joined by Mellon Intergenerational Scholars, former interns themselves, for the summer internship with partners at the Tijuana River National Estuarine Research Reserve (TRNERR) in the US and *Colonos de Divina Providencia AC* (CDP) in Tijuana.

Linked by flows of water and waste, the western canyons of Tijuana and the southern San Diego County communities of San Ysidro and Imperial Beach are interrupted by the growing border wall infrastructure between the US and Mexico. Water, air, ocean currents, and flows of waste, however, do not observe the physical wall. The Divina Community Station is focused on shared goals of waste reduction, ecological resilience, and binational community-building. The central task of the summer interns is to strengthen the capacity to understand and act in the canyon communities while advancing long-term research goals.

Our goal at BSFI is to help students understand the complicated political and ecological landscape of the border communities through working directly in the Tijuana Estuary in the US and the peripheral canyon communities of Tijuana. This goal would not be attainable without the support and energy of our two partners:



- *Colonos de Divina Providencia, AC*, in particular Presidenta Rebeca Ramirez Robles and *promotoras* Maria de la Luz Alejandre and Graciela Vargas.



- The Tijuana River National Estuarine Research Reserve, in particular Kristen Goodrich and Ana Xochilt Eguiarte from the Coastal Training program.

Student projects emerge from an ongoing relationship between faculty and staff at UCSD and our partners, and are conducted by a mixed team from each organization with extensive experience working in the binational watershed.

2019 Interns:

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| • Armando Olea | UCLA Latinx Studies |
| • Daniel Ochoa | UCSD Earth Systems |
| • Natalie Muñoz | UCSD Sociology / International Studies |
| • Malka Williams | UCSD Urban Studies and Planning |
| • Maria Montes Arvizu | UCSD Speculative Design |

Teaching Staff:

- | | |
|------------------------|--|
| • Kyle Haines | Field Coordinator, UCSD Cross-Border Community Station |
| • Ana Eguiarte | COLEF Environmental Administration, TRNERR |
| • Kristen Goodrich | UCI Social Ecology, TRNERR |
| • German Partida | Mellon Intergenerational Scholar |
| • Nadia Lopez | Mellon Intergenerational Scholar |
| • Paul Moscoso Ríofrio | Estudio Cruz and Forman |
| • Daniela Perez | UCSD Center on Global Justice |
| • Teddy Cruz | Project PI, Director UCSD Cross-Border Initiative |
| • Fonna Forman | Project PI, Director UCSD Cross-Border Initiative |

Volunteers:

- | | |
|------------------------|--|
| • Harold Ruben | |
| • Alexis Chavez | UCSD Alumni, Former Intern |
| • Magnolia Garcia | UCSD Alumni, Former Mellon Intergenerational Scholar |
| • Francesca Hummler | UCSD Media and Biochemistry |
| • Farshid Barzamadegan | UCSD Visual Arts |
| • Ella McDougal | MIIS, TRNERR Intern |



Final day group shot with (L-R): Nadia López, Magnolia García, Germán Partida, Maria Montes, Daniel Ochoa, Kyle Haines, Kristen Goodrich, Rebeca Ramírez, Natalie Muñoz, Armando Olea, and Malka Williams.

Group Foci:

This summer interns worked together to produce workshops ranging from a video project with local students, planting milkweed for native butterflies with the *tercera edad* (retirement) group, and a community workshop on the everyday impacts of trash and some political and biological strategies for building resilience to flooding and erosion. Students also worked in smaller groups to 1) assist a local native plants nursery, *Vivero Hormiguitas*, and 2) to deepen community capacity in Divina Providencia by advancing online systems for reporting dumping, building social media presence and working with local youth to develop pedagogical gardens for food and native plants at the Divina Community Center. These groups delivered activities and games to explain the relevance of their work at the final community workshop, the details of which can be found below.

a. Deepening Community Engagement in Divina



L: Daniela Perez leads in a stretching and breathing exercise. R: Local kids paint their handprints on the Divina Community Center.

Interns placed in the group working exclusively at the Community Station in Divina Providencia worked to increase the capacity of CDP in several ways. CDP is focused on helping the most vulnerable members of their community, providing a subsidized meals program, support for elderly residents, and free medical care on a weekly basis.

Interns were asked to do a variety of tasks related to increasing the social media presence (through building a facebook profile, posting updates, and recruiting locals interested in helping or attending events), as well as working directly with local children and elderly folks that use the Divina Community Center. This included a day of activities with the children, delivering a short lesson on butterflies followed by planting milkweed with the *tercera edad* group, and working with CDP *promotoras* to improve design of the pedagogical garden created in the Community Center patio.



L: Monarch butterfly taller chalkboard artwork. C: A small native milkweed seedling planted at the butterfly taller. R: Students give some basic information about monarchs.

Finally, interns working in Divina were asked to develop a large-format matching game for the final workshop based on understanding individual, collective, and institutional actions for dealing with issues of trash and pollution in the canyon communities. This game incorporated work on developing online forms for filing *denuncias* (formal complaints), organizing community clean ups, and growing an online and personal community to increase capacity for collective action.

b. Understanding Native Plants with Hormiguitas Vivero



L: Jorge Ibañez telling stories about *Hormiguitas Vivero*. R: Hormiguitas greenhouse growing *jojoba*.

Interns placed in the second group made weekly visits to the native plant nursery in neighboring Las Flores to work with Jorge Ibañez and Adela Bonilla. Interns here did some physical support work at the site, learned about the variety of native plants and how to propagate and care for them

in the nursery, and mapped buildings and pathways of the physical site. In order to help increase the capacity of the nursery, interns created an inventory of current plants, translating key facts about each species from an English language resource, researched and recommended new native species for expanding the variety offered at the nursery, and developed poster-size materials for basic information and plant inventory/price for the *vivero* to use in the future.



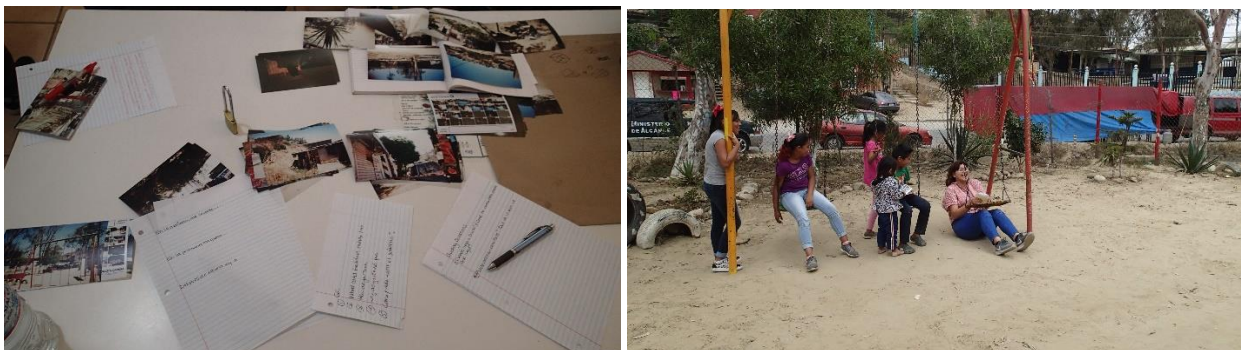
L: Ella, Daniel, and Maria learning to unshuck Lemonadeberry (*saladito*) seeds. R: Before and after shucking.

Interns spending time at *Hormiguitas* were asked to present their understanding of the role of native plants in controlling erosion in the final workshop. To do so they constructed two demonstrations with pots of soil with different coverings. After explaining some of the social and ecological benefits of cultivating native plants they invited local kids to come and pour water over a) two tubs, one with plants and one without and b) four tubs with no cover, leaves, gravel, and trash on top, then observe and comment to the group of adults the differences they noted.

Full Group Projects:

Video Taller

The video workshop project brought kids from the local community into four weekly meetings to co-create a video about trash in the community to show their parents and grandparents at the final workshop. Interns were paired with local students and tasked with understanding one key area each in the flow of waste from the top of the canyon to the ocean at the estuary mouth. Key efforts were provided by Farshid Barzamandegan, Daniela Pérez, Jona Maier, Marcello Maltagliatti, and Dario Guazzo.



L: Disposable camera pictures and text writing. R: Swing accidents on film day.

First, Daniela Pérez conducted a lesson on storytelling which led to local/intern pairs walking through the community and taking pictures together on disposable cameras. Next, interns worked

with their local youth partners to create short narratives based on their pictures about the effects of black water, trash burning, and flooding caused by choked canals on their everyday life. Based on these narratives, students interviewed elderly residents about their hopes for the future and ideas to resolve the trash problem.



L: Armando and Alexis filming around Divina Providencia. R: Jona showing kids how the camera works.

The narratives and conversations produced were filmed by Farshid and Dario, and edited by Jona and Marcello. In a final day of exercises, each of the interns brought an activity for their local youth partners, from painting rocks in the garden to hitting a piñata, and then presented them with new backpacks, school supplies, and a certificate of accomplishment in front of their parents. During the final workshop, youth partners were recognized again and introduced their video to the community, before participating in the workshop activities.

Community Workshop on Trash and Erosion

The final project for the summer served both to link together the different foci of the group and pilot a workshop on managing marine debris for our partners at TRNERR. Approaching the problem of trash in the canyon communities, this workshop was intentionally light on lecture and heavy on demonstration and activity. Geared to be interesting to kids and adults, it took place in just over an hour, and included both showing the video produced by kids and interns and doing activities based on work at the community center and in the *vivero* (i.e. focused on native plants and collective action).



L: Delivering the erosion demonstration at the final workshop. R: The matching game at the final workshop.

Following the video, Ana Eguiarte from TRNERR talked briefly about the consequences of trash and pollution on the health of humans and ecosystems, after which the interns conducted their demonstrations and game, detailed in brief below:

a. Erosion Demos 1 & 2 (Daniel and Maria)

Each demo consisted of several boxes filled with soil side-by-side. Interns drilled holes in each box and fitted them with washers to allow for water to flow out of the side below the soil-line. These models were set up as follows:

- *Model #1* = 2 27-gallon boxes with plants in one, no plants in the other to show how plants filter water and secure loose soils.
- *Model #2* = 4 8-gallon boxes with different covers (bare, leaves, gravel, trash) to show

Clear pitchers were placed beneath each of these spouts to capture water for comparison. Interns, with youth participants, poured equal amounts of water over each box, allowed the water to drain, then asked the kids to describe what they saw.



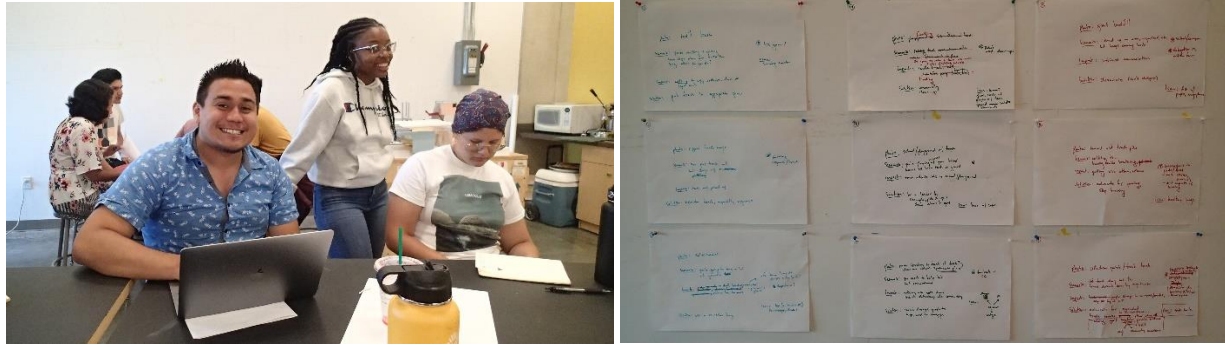
L: Drilling holes and sealing. R: Planting and testing the demos.

b. Matching Game (Armando, Malka, Natalie)

The basic point of the matching game was to organize some of the outstanding issues in the canyon communities by the level of collective action necessary, from individual to collective to institutional scales. The game itself consisted of matching actions to pictures of negative outcomes taken from specific places around the canyon and Divina Providencia in particular. Working together, interns designed a game that consisted of:

- 9 squares with photos of bad outcomes (organized loosely by scale)
- 9 actions (words + icons) to match with bad outcome squares
- 9 scenarios + extra info (facts about trash, etc)

At the final workshop, interns presented situations that represented everyday issues related to the pictures mounted in a 3x3 grid on the wall in the community center. First reading a short narrative about how that issue presented itself in Divina, interns then selected from the youths in the audience to choose an associated action and place that action (via Velcro) onto the picture to which it corresponds. While a little difficult to conduct with a large group, this exercise proved interesting and useful, engaging parents and grandparents in the back who shouted answers as well as the young kids directly involved in the game itself.



L: Working on workshop materials in the studio at UCSD. R: Mock-up of matching game.

Thank You!

This internship is not possible without the generous help of many people on both sides of the border. Beyond our partners at UCSD, TRNERR, and CDP, we would like to acknowledge the help and participation of the following folks from Tijuana who generously shared their time and experience working on regional issues with our group.

Gracias a...

Jorge Ibañez, Vivero Hormiguitas

Adela Bonilla

Samuel Perez, Ecoparque Urban Agriculture

Margarita Díaz, Proyecto Fronterizo de Educación Ambiental